La Scuola d'Italia / Dr. Huson / IB Math Name:
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5.8 Classwork: Calculus and Statistics (no calculator)

1. Let $f(x) = x^3 - 3x + 2$.

   (a) Find $f'(x)$. [1]

   (b) Find the gradient of the graph of f at $x = 2$. [2]

   (c) Find the equation of the tangent line to the graph of f at $x = 2$. Give your answer in the form $y = mx + c$. [3]

   (d) Find the coordinates of the stationary points of f. [2]

2. Let $f(x) = 3x^2 - 6x$.

   (a) Find $\displaystyle\int (3x^2 - 6x)\,\mathrm{d}x$. [2]

   (b) Show that $f(x) = 0$ when $x = 0$ and $x = 2$. [1]

   (c) Find the area of the region enclosed by the graph of f and the x-axis, between $x = 0$ and $x = 2$. [3]

   (d) A function g has derivative $g'(x) = 3x^2 - 6x$ and $g(1) = 5$. Find $g(x)$. [2]

3. A manufacturer produces items. Each item is independently classified as defective or non-defective.

   (a) In a random sample of 3 items, the probability that any one item is defective is $\frac{1}{3}$. Let X be the number of defective items.

      (i) Write down the distribution of X. [1]

      (ii) Find $\mathrm{P}(X \leq 1)$. [3]

   (b) The mass of items follows a normal distribution with mean 50 g and standard deviation σ g. It is known that 2.28% of items have mass greater than 56 g.

      You are given that $\mathrm{P}(Z > 2.00) = 0.0228$.

      (i) Find the value of σ. [2]

      (ii) Find $\mathrm{P}(\text{mass} < 44)$. [2]

4. Let $f(x) = x^3 - 6x^2 + 9x + 1$.

   (a) Find $f'(x)$. [1]

   (b) Find the x-coordinates of the stationary points. [2]

   (c) Use the second derivative to determine the nature of each stationary point. [3]

   (d) Find the y-coordinates and hence state the local maximum and local minimum of f. [2]